

CLEARCOVE POOLS

The ClearCove Pool Owner's Handbook

The Complete Guide to Water Chemistry, Seasonal Care, and Keeping Your Pool Crystal Clear Without Guessing

FREE HOMEOWNER GUIDE · 2026 EDITION
ClearCove Pools

clearcovepools.com

How to Use This Book

You don't need to read this cover to cover before your next test strip. Most people pick this up for one of three reasons: you just inherited a pool with a new house and nobody explained how to run it, you woke up to a green swamp and every forum you've read says something different, or you're tired of guessing "a cup of this, a splash of that" and want to actually understand what you're doing.

Whatever brought you here, use the book that way. If your pool is green right now, skip to Chapter 6. If you're opening for the season this week, go straight to Chapter 8. If you just want the short version of what to test and when, Chapter 2 is your calendar and your 10-minute weekly routine.

But if you have an evening to spare, read it in order once. Pool chemistry is not complicated once you understand the handful of relationships that make it work, and every chapter after Chapter 1 leans on that foundation. The pool owners who stop fighting their water aren't the ones who buy the most chemicals. They're the ones who understood the relationship between chlorine, pH, alkalinity, and stabilizer well enough to fix the actual cause instead of throwing another jug of shock at a symptom.

One more thing. Every dosing example in this book is based on standard industry dosing math for an average pool, not a guarantee for your specific pool. Pool size, surface material, sun exposure, bather load, and your specific test kit all affect real-world results. Treat this as the framework that helps you understand what your test numbers mean and why, and always verify with your own test kit before and after any significant chemical addition.

Before You Add Anything: Chemical Safety

Pool chemicals are effective precisely because they're reactive, and that means they deserve real respect. CDC surveillance has tracked an estimated 13,500+ emergency-department visits a year in the US tied to pool chemical injuries, most from breathing fumes, and more than a third involving people under 18.

- **Never mix pool chemicals directly with each other** — especially chlorine products and acid. That reaction releases toxic chlorine gas. Add each chemical to the pool water separately, never chemical-to-chemical, and never into the same bucket or scoop without rinsing it first.
- **Add acid to water, not water to acid**, when diluting, and add it slowly, away from your face, with the pump running.
- **Wear gloves and eye protection** when handling concentrated chemicals, and work in a ventilated area, never a closed shed with no airflow.
- **Store chemicals in their original, labeled containers** in a cool, dry, ventilated space, separated from each other (acids kept apart from chlorine products), and out of reach of children and pets.
- **If a chemical splashes in your eyes**, rinse with clean water for at least 15–20 minutes immediately, then contact Poison Control (1-800-222-1222 in the US) or seek emergency care if stinging continues.

- **If you spill a chemical**, don't rinse it down a drain. Contain it with an inert absorbent (never sawdust on an oxidizer). If a spill produces fumes or you can't safely contain it, clear the area and call for help rather than attempting cleanup yourself.

This book is educational information, not a Safety Data Sheet (SDS). Always follow the specific product's label instructions and SDS, which take precedence over anything general written here.

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Chapter 1: How Pool Water Chemistry Actually Works

A swimming pool is a large volume of standing water that sunlight, bather load, dirt, and organic material are constantly working to contaminate, and that you're constantly working to keep clear, balanced, and safe to swim in. Unlike a septic system or a well, there's no soil or aquifer doing background work for you. Every bit of the water's chemistry is something you're actively managing, which is exactly why understanding the relationships between the numbers matters so much more than memorizing a shopping list of chemicals.

The two jobs pool chemistry does

Pool water chemistry does two separate jobs at once, and most owner confusion comes from mixing them up. **Sanitization** is about killing bacteria, algae, and pathogens — primarily chlorine's job. **Balance** is about protecting your pool's surfaces, equipment, and swimmers from water that's too aggressive (corrosive) or too passive (scale-forming) — primarily the job of pH, alkalinity, and calcium hardness. A pool can have plenty of chlorine and still be poorly balanced, corroding equipment and irritating swimmers' eyes. A pool can be perfectly balanced and still turn green if chlorine runs out. You need both working correctly, and they interact with each other, which is why Chapter 5 walks through the correct order for adjusting them.

Free chlorine vs. combined chlorine

This is the single most important distinction in pool chemistry, and the one most owners never learn. **Free chlorine (FC)** is the chlorine that's actually available to sanitize — the active, working portion.

Combined chlorine (CC) is chlorine that's already reacted with contaminants (sweat, sunscreen, urine, organic material) and formed chloramines — a used-up, mostly ineffective form that's also the primary source of that harsh "chlorine smell" and eye irritation people blame on "too much chlorine." A properly chlorinated pool with a strong chlorine smell almost always has too much combined chlorine and not enough free chlorine, not the other way around.

A test kit that only measures "total chlorine" (FC + CC combined) doesn't tell you this critical difference. Use a test kit or test strips that report free chlorine specifically, ideally alongside total chlorine so you can calculate combined chlorine (total minus free) and know whether you're dealing with a sanitizing problem or a chloramine problem.

pH: the water's acid/base balance

pH measures how acidic or basic your water is, on a scale where 7.0 is neutral. CDC and the Model Aquatic Health Code (MAHC) put the operational target for residential pools at **pH 7.0–7.8**, with 7.4–7.6 generally considered the comfort sweet spot closest to the pH of human eyes and mucous membranes, which is part of why well-balanced pool water doesn't sting. pH drifts constantly from bather load, rain, sunlight breaking down chlorine, and the specific chemicals you add, which is why it needs regular testing and adjustment rather than a "set it and forget it" approach.

Total alkalinity: pH's shock absorber

Total alkalinity (TA) measures the water's resistance to pH change — essentially a buffer. Target range is typically 80–120 ppm. Low alkalinity lets pH swing wildly with small additions of chemicals or even normal bather use, a frustrating and hard-to-manage pattern often described as "pH bounce." High alkalinity makes pH sluggish and hard to adjust, and can push pH upward over time on its own. Getting alkalinity into range first is why Chapter 5 puts it early in the balancing sequence: fix the shock absorber before you fine-tune the thing it's absorbing shocks for.

Cyanuric acid (CYA): chlorine's sunscreen

Cyanuric acid, also called stabilizer or conditioner, protects free chlorine from being rapidly broken down by UV sunlight. Without adequate CYA, an outdoor pool can lose most of its chlorine to sunlight within a few hours on a sunny day. But CYA has a cost: higher CYA levels reduce chlorine's effective killing power at a given free chlorine reading, meaning a pool with high CYA needs a higher FC number to achieve the same actual sanitizing strength as a pool with lower CYA. This relationship, often summarized loosely as needing to maintain FC at roughly 7.5% of your CYA level as a practical minimum, is one of the most misunderstood concepts in pool care and a major reason algae blooms happen even when a test strip shows "chlorine present."

Calcium hardness: protecting surfaces and equipment

Calcium hardness measures dissolved calcium in the water. Too low, and water becomes aggressive toward plaster, grout, and metal equipment, pulling calcium out of surfaces to compensate (etching). Too high, and calcium can precipitate out as scale on surfaces and equipment, particularly heaters. Target range depends on pool surface type: roughly 175–225 ppm for vinyl and fiberglass pools, and 200–275 ppm (up to 400 ppm) for concrete/plaster pools, since plaster has exposed calcium-based surfaces to protect that vinyl and fiberglass don't.

The mental model that ties it together

Picture pool chemistry as four dials that all affect each other: chlorine kills what needs killing, pH determines how comfortable and effective that chlorine actually is, alkalinity keeps pH from swinging wildly every time something changes, and CYA determines how much chlorine you actually need to maintain to keep up with sunlight. Adjust one dial in isolation without understanding its effect on the others, and you'll chase your tail. Understand the relationships, and pool care becomes a short, predictable weekly routine instead of a mystery.

The bottom line: Pool chemistry does two separate jobs — sanitizing (chlorine) and balancing (pH, alkalinity, calcium) — and cyanuric acid determines how much chlorine you actually need to do the sanitizing job effectively. Free chlorine, not total chlorine, is the number that tells you whether your pool is actually protected, and understanding how these numbers affect each other is what separates confident pool ownership from constant guesswork.

Chapter 2: The Testing & Maintenance Calendar — Your 10-Minute Weekly Routine

This is the chapter to tape inside a cabinet near your pool records. Test twice a week, brush once a week, shock when it counts — that's the whole system, and most weeks it takes about 10 minutes.

Your weekly rhythm

When	Task	Why
2–3x/ week	Test free chlorine (FC) & pH	Catches drift before it becomes a problem; daily during heavy use or a heat wave
Weekly	Test total alkalinity (TA); skim, brush walls, empty skimmer & pump baskets	TA keeps pH stable; debris removal prevents algae and clogs
Monthly	Test cyanuric acid (CYA) and calcium hardness; check filter pressure vs. clean baseline	CYA affects how much chlorine you actually need; rising pressure means it's time to backwash/clean

Do / Don't

Do

- Add one chemical at a time and let it circulate before testing again
- Add acid slowly, in a stream, with the pump running
- Store chemicals in a cool, dry, separate space
- Measure your pool's actual gallons before dosing

Don't

- Never mix chemicals together — especially chlorine and acid
- Don't add chemicals to a stagnant pool with the pump off
- Don't guess dosages from memory or a forum comment

How often to test, in detail

Parameter	Testing frequency (swim season)
Free chlorine (FC) and pH	2–3 times per week minimum; daily during heavy use or hot weather
Total alkalinity (TA)	Weekly
Cyanuric acid (CYA)	Monthly, or after adding stabilizer/stabilized chlorine
Calcium hardness	Monthly

Parameter	Testing frequency (swim season)
Total dissolved solids (TDS)	A few times per season, or if water seems consistently hard to balance
Phosphates	Optional, a few times per season if algae has been a recurring problem

A reasonable weekly routine

1. **Test FC, CC, and pH** (2–3 times per week, more during peak swim season or a heat wave).
2. **Add chlorine as needed** to maintain your target free chlorine range for your specific CYA level (Chapters 1 and 3).
3. **Brush pool walls and floor** at least once a week, more for pools prone to algae, since brushing physically disrupts algae before it can establish and helps chlorine reach surfaces effectively.
4. **Skim debris and empty skimmer baskets** as needed, ideally every visit.
5. **Vacuum** as needed based on visible debris — weekly is typical for most residential pools during the season.
6. **Backwash or clean the filter** per your filter type's schedule (Chapter 10).
7. **Check equipment** (pump running quietly, no unusual noises, proper water flow) during your regular visit.

Monthly tasks

- Test and adjust CYA, calcium hardness, and TDS as needed
- Inspect pool surfaces for any signs of scale, etching, or staining developing
- Deep-clean the filter per manufacturer guidance if not already part of your regular cleaning cycle

What to log

- Every test result with date (FC, CC, pH, TA, CYA at minimum)
- Every chemical addition: what, how much, and when
- Weather and usage notes (heavy rain, a pool party, extended heat) that help explain unusual readings
- Equipment maintenance and any repairs

Appendix A has a ready-to-use log template. A consistent log turns pool care from a reactive scramble into a pattern you can predict and manage, and it's invaluable for diagnosing a problem when something suddenly goes wrong.

The bottom line: Test free chlorine and pH several times a week during swim season, alkalinity weekly, and CYA and calcium monthly. Brushing and skimming are as important as chemical testing for preventing algae and keeping water genuinely clear, and a simple log turns scattered test results into a pattern you can actually manage.

Chapter 3: The Core Numbers — Target Ranges and What They Mean

Standard target ranges

Parameter	Target range	Notes
Free chlorine (FC)	1–4 ppm	CDC/MAHC operational range for residential pools; CDC's safety floor is at least 1 ppm, at least 2 ppm if cyanuric acid is used. Run toward the higher end when CYA is high or bather load is heavy.
Combined chlorine (CC)	Below 0.5 ppm	Above this, shock is typically needed (Chapter 4)
pH	7.0–7.8	CDC/MAHC operational range; 7.4–7.6 is often considered the comfort sweet spot
Total alkalinity (TA)	80–120 ppm	Lower end for plaster pools sometimes preferred; check your specific surface guidance
Cyanuric acid (CYA)	30–50 ppm (traditional chlorine); 60–80 ppm (salt systems)	Higher isn't automatically better; very high CYA (100+ ppm) causes chlorine lock and may require partial water replacement to correct
Calcium hardness	175–225 ppm (vinyl/fiberglass); 200–275 ppm, up to 400 ppm (plaster)	Too low risks surface etching; too high risks scale
Total dissolved solids (TDS)	Below roughly 1,500–2,000 ppm above the fill-water baseline	Rising TDS over a season is normal; very high TDS can make water harder to balance

The FC-to-CYA relationship in practice

A widely used practical guideline: maintain free chlorine at roughly 7.5% of your CYA reading as a working minimum. At 30 ppm CYA, that's roughly 2.25 ppm FC minimum. At 60 ppm CYA, that's roughly 4.5 ppm FC minimum. This is why two pools can both show "chlorine present" on a basic test strip and behave completely differently: the one with high CYA and FC sitting near the low end of the 1–4 ppm range may functionally have almost no effective sanitizing power, while the one with low CYA and the same FC reading is well protected.

Why pH matters more than a pass/fail line

pH doesn't just affect comfort — it directly changes how much of your chlorine is actually working. Per the National Swimming Pool Foundation's Pool and Spa Operator Handbook, the active, killing form of chlorine (hypochlorous acid) makes up roughly 76% of total chlorine at pH 7.0, drops to about 66% at pH 7.2, about 50% at pH 7.5, and just 33% at pH 7.8. In practical terms, a pool testing pH 7.8 has roughly

half the effective sanitizing power of the same ppm reading at pH 7.2. That's why professionals generally correct pH into the 7.2–7.6 range before shocking, so the dose isn't partly wasted.

Why "my chlorine keeps disappearing" often isn't really a chlorine problem

Owners frequently interpret rapidly dropping FC as "I need to add more chlorine, and more often." Sometimes that's the right answer (heavy bather load, hot sunny weather, low CYA letting sunlight burn through chlorine fast). But persistently fast-dropping chlorine with normal usage and adequate CYA can also indicate an early, not-yet-visible algae bloom consuming chlorine, or a high chlorine demand from accumulated organic material that a shock treatment (Chapter 4) needs to catch up on, rather than routine daily dosing.

Reading your numbers together, an example

Say your test shows: FC 1.0 ppm, CC 0.8 ppm, pH 7.1, TA 60 ppm, CYA 80 ppm.

Reading this together: your CYA is high enough that 1.0 ppm FC is providing very little real sanitizing power (well under the 7.5% guideline of roughly 6 ppm minimum at this CYA level). Your combined chlorine at 0.8 ppm is elevated, suggesting the existing chlorine is mostly used up fighting contaminants rather than sanitizing fresh. Your pH is on the low end and your alkalinity is below range, which together suggest the water may feel slightly aggressive and will be prone to pH bounce until alkalinity is corrected. The correct response isn't "add a little chlorine": it's raise alkalinity first, then shock to clear the combined chlorine and catch up FC to a level appropriate for that CYA (Chapters 4 and 5 walk through this sequence).

The bottom line: Target ranges are a starting point, but the real skill is reading the numbers together, especially the FC-to-CYA and pH-to-chlorine-strength relationships, rather than treating each one in isolation. A single low or high number rarely tells the whole story; the pattern across all of them does.

Chapter 4: Chlorine, Shock, and Sanitizing Options

Types of chlorine

- **Liquid chlorine (sodium hypochlorite):** no CYA added, doesn't raise pH much when properly diluted commercial-strength, widely used for both daily dosing and shocking. Sold at 10% or 12.5% concentration; degrades fast in storage, especially in heat — a 12.5% jug loses roughly half its strength in about 180 days at 77°F, but closer to 48 days at 95°F, so buy only what you'll use within about a month in hot climates.
- **Cal-hypo (calcium hypochlorite):** strong, no CYA added, but raises calcium hardness with each addition, which matters over a season of repeated use, particularly for pools already on the higher end of calcium hardness. Best suited to plaster/gunite/concrete pools; on vinyl liners, undissolved granules that settle before fully dissolving can bleach or chemically burn the material.
- **Dichlor:** granular, contains CYA, convenient for daily dosing early in the season when CYA needs building up, but repeated use over a season can push CYA higher than desired since every dose adds more stabilizer.
- **Trichlor (tablets/pucks):** contains a high proportion of CYA per dose and is quite acidic, commonly used in floating or in-line feeders for slow, steady dosing. Convenient, but the CYA contribution adds up over a season and can push levels high if used as the sole chlorine source for extended periods without monitoring CYA. Sealed, stored cool and dry, trichlor tablets hold roughly 98% of their available chlorine at the 3-year mark and can remain viable 5+ years unopened.

Choosing between them

Many pool owners use a combination: a stabilized source (trichlor or dichlor) for convenient baseline dosing when CYA is still building, switching toward unstabilized liquid or cal-hypo once CYA reaches target range, to avoid pushing it higher than needed. Track your CYA level specifically if you're using any stabilized chlorine product regularly, since it's the one parameter that only goes up from these products (short of dilution/partial drain) and can quietly climb over a season without you noticing until chlorine seems to "stop working."

Shocking: what it actually does

Shocking means adding a large dose of chlorine (or a non-chlorine oxidizer) to rapidly raise free chlorine to a level high enough to break down combined chlorine (chloramines), oxidize accumulated organic material, and, when done at a high enough level, kill algae. It's a maintenance tool for catching the water up, not a component of your everyday dosing routine.

When to shock

- Weekly or biweekly as routine maintenance for many owners, particularly during heavy-use periods

- Whenever combined chlorine exceeds roughly 0.5 ppm
- After heavy bather load (a pool party, a hot crowded weekend)
- After any sign of algae (Chapter 6)
- After a heavy rainstorm, which can introduce contaminants and dilute chlorine
- At pool opening, to catch up water that's been sitting since closing (Chapter 8)

Non-chlorine shock (potassium monopersulfate / MPS)

An oxidizer that clears combined chlorine and organic material without a chlorine spike, useful when you want to oxidize the water but don't want to wait out an elevated chlorine level before swimming again — many non-chlorine shock products allow swimming again within 10–15 minutes, far shorter than chlorine shock's waiting window. Standard maintenance dose runs about 1 lb per 10,000 gallons, with a heavier shock-level dose around 2 lb per 10,000 gallons for tougher contamination. It doesn't have algaecidal properties at typical doses the way a proper chlorine shock does, so it's not a substitute for chlorine shock when actual algae is present.

A note on "shock and swim" timing

Always follow the specific product's swim-safe waiting guidance rather than assuming a fixed universal number; different products and different shock levels have different safe-to-swim thresholds, generally tied to free chlorine dropping back to a safe range (commonly below roughly 5 ppm for many jurisdictions/guidelines, though check your specific product and local guidance).

The bottom line: Different chlorine types trade off convenience against their CYA contribution; track CYA specifically if you rely on stabilized products all season. Shocking is a periodic maintenance tool for catching water up (clearing combined chlorine, organic material, and algae), not a replacement for consistent daily/weekly chlorine dosing.

Chapter 5: Balancing Water — The Correct Order of Operations

Getting balance chemicals in the wrong order wastes money and time, because each parameter affects the others. This is the sequence that avoids fighting your own adjustments.

The correct sequence

- 1. Total alkalinity first.** Because alkalinity buffers pH, adjusting it first means your subsequent pH adjustment will actually hold instead of drifting again within a day. Raise alkalinity with sodium bicarbonate (baking soda); lower it with muriatic acid added carefully, typically in the deep end with circulation running, since the process to lower alkalinity specifically also affects pH and requires a particular technique (adding acid in one spot without circulating immediately, sometimes called the "acid spot" method, to target alkalinity reduction with less pH impact); follow specific product guidance.
- 2. pH second.** Once alkalinity is in range, adjust pH: raise with soda ash (sodium carbonate); lower with muriatic acid (standard pool-grade strength is 31.45% hydrochloric acid) or dry acid (sodium bisulfate). Small, incremental adjustments are safer than one large dose, since overcorrecting pH is common and easy to do.
- 3. Calcium hardness third.** Adjust with calcium chloride to raise; lowering calcium hardness generally requires partial drain-and-refill with lower-hardness water, since there's no simple chemical to remove dissolved calcium.
- 4. Cyanuric acid fourth,** if needed, using a stabilizer product; roughly 1 lb of granular stabilizer per 10,000 gallons raises CYA by approximately 10–13 ppm depending on the product. There's no simple chemical to lower CYA meaningfully, only dilution through partial drain-and-refill.
- 5. Chlorine last.** Once the water is properly balanced, chlorine works more efficiently and predictably, and you're not fighting pH swings while trying to also manage sanitizing.

Why the order matters, in practice

Adding chlorine to badly out-of-balance water isn't dangerous, but it's inefficient: acidic, low-alkalinity water burns through chlorine faster and irritates swimmers; high-pH water reduces chlorine's killing effectiveness even at a seemingly adequate FC reading (Chapter 3). Balancing first, then dosing chlorine, means every dose of chlorine you add afterward is working near its full potential instead of being partially wasted fighting bad water chemistry.

Dosing math: the honest caveat

Every dosing chart and calculator (including the free one on [ClearCovePools.com](https://clearcovepools.com)) works from your pool's volume and standard industry dosing tables, giving you a solid starting estimate, not a guaranteed final answer. Real-world results vary with your actual water chemistry, pool surface, and the specific

product's concentration. Always retest after any significant addition and adjust incrementally rather than dumping in a large calculated dose and walking away for a week.

A worked example: 10,000 gallons

Say a 10,000-gallon pool tests: TA 60 ppm (target 80–120), pH 7.1, calcium 90 ppm (target 200+ for a plaster pool), FC 0.5 ppm (target 1–4, adjusted for CYA).

1. **Raise alkalinity:** baking soda raises TA by roughly 10 ppm per 1.5 lb per 10,000 gallons on typical product labels. To move from 60 to 100 ppm (a 40 ppm rise) needs roughly 6 lb, added in stages with the pump running. Retest in a few hours to a day.
2. **Adjust pH:** likely needs only a modest boost, since raising alkalinity often nudges pH up somewhat on its own — retest before assuming you need additional soda ash.
3. **Raise calcium hardness:** calcium chloride dosing varies by product concentration; follow the label's ppm-per-pound-per-10,000-gallons rate to move from 90 toward the 200–275 ppm plaster target, added gradually and retested between doses.
4. **Dose chlorine last:** once balanced, bring FC into the 1–4 ppm range appropriate for the pool's CYA level.

Each step gets a retest before moving to the next, rather than dumping everything in at once and hoping it settles into place.

A worked example: 20,000 gallons, cyanuric acid

A 20,000-gallon pool tests CYA at 20 ppm and the owner wants to bring it to 40 ppm (a 20 ppm rise) ahead of summer. Using the approximate rate of roughly 1 lb of granular stabilizer per 10,000 gallons raising CYA by about 10–13 ppm, a 20,000-gallon pool needing a 20 ppm rise would use roughly 3–4 lb total, split across the pool's two 10,000-gallon "units" and dissolved via a skimmer sock since granular stabilizer dissolves slowly. Always confirm the exact rate against the specific product label, since formulations vary.

The bottom line: Balance in this order: alkalinity, then pH, then calcium hardness, then CYA if needed, then chlorine. Adjusting chlorine before the water is balanced wastes chlorine and money. Retest between each step rather than adding everything at once, since each adjustment affects the others.

Chapter 6: Algae — Prevention, Types, and Recovery

Why algae happens

Algae spores are airborne and constantly entering pool water; they're always present in small amounts. Algae blooms happen when conditions favor their growth faster than your chlorine and filtration can suppress them: insufficient free chlorine relative to CYA (Chapter 3), poor circulation or a dead spot in the pool, inadequate brushing letting algae establish on surfaces where chlorinated water doesn't reach as effectively, and warm, sunny conditions that accelerate growth.

The most common cause: chronic low free chlorine relative to CYA

Far more algae blooms trace back to a slow, chronic FC deficit (often from high CYA quietly reducing chlorine's effective strength, as covered in Chapter 3) than to any single dramatic event. This is why understanding and maintaining the FC-to-CYA relationship is the single highest-leverage algae prevention habit in this entire book.

Types of algae

- **Green algae:** the most common type, turns water green to varying degrees from a slight tint to a full swamp; generally the easiest to kill with adequate shock treatment once caught.
- **Mustard/yellow algae:** a yellowish, often patchy growth on shaded walls; more chlorine-resistant than green algae and prone to returning if not fully eradicated, sometimes requiring a specific algaecide alongside shock. Polyquat 60 is a common non-metallic choice, dosed at roughly 2–4 fl oz per 10,000 gallons for routine maintenance and 11–17 fl oz per 10,000 gallons for visible algae, following the product label.
- **Black algae:** technically cyanobacteria, not true algae. It roots into plaster/grout and forms a protective outer layer, which is why routine chlorine and standard algaecide don't clear it. Treatment typically means raising free chlorine to roughly 10–20 ppm and holding it elevated for about 24 hours while brushing repeatedly with a stainless/wire brush on plaster/gunite/concrete (soft nylon on vinyl/pebble/quartz) to strip the protective layer before chemical treatment can penetrate. Full clearing commonly takes 1–3 weeks of daily brushing.
- **Pink/red slime:** technically a bacteria rather than true algae, but grouped here since owners encounter and treat it similarly; often found in corners, steps, and around fittings; typically requires thorough cleaning combined with shock.

Recovery process for a green pool

1. **Remove large debris** (skim leaves and obvious debris before chemical treatment).
2. **Test what you can** (pH and alkalinity, if the water is clear enough to get any reading; heavily green water may not give reliable chlorine readings until you begin treatment).

3. **Balance alkalinity and pH first** if testable, since chlorine works far more efficiently once the water is in a reasonable range, following Chapter 5's sequence as far as you're able given limited visibility.
4. **Shock aggressively**, typically at a significantly higher dose than routine maintenance shock, following the product's specific algae-treatment guidance for your pool volume; this is not the moment for a light routine dose.
5. **Run the filter continuously** (24 hours a day) during recovery rather than a normal partial daily schedule, to keep debris and dead algae moving toward the filter.
6. **Brush daily** during recovery to keep dead algae from resettling on surfaces and to help chlorine reach every area.
7. **Retest and re-shock as needed** until FC holds steady rather than crashing to zero within hours, which indicates the chlorine demand from algae and organic material is finally being met.
8. **Vacuum out settled dead algae** once the water clears enough to see the bottom, ideally to waste (bypassing the filter) if your system supports it, since large amounts of dead algae can overwhelm a standard filter cycle.
9. **Clean or backwash the filter thoroughly** once recovery is complete, since a filter used through an algae recovery accumulates a heavy load.
10. **Retest fully** (FC, CC, pH, TA, CYA) once water is clear and stays clear over 24–48 hours, and resume normal maintenance dosing from a properly balanced baseline.

Preventing recurrence

- Maintain FC appropriately for your CYA level consistently, not just when the water looks fine
- Brush weekly, more often in algae-prone areas (shaded corners, steps, behind ladders)
- Keep circulation running long enough daily to turn over the pool's full volume (check your pump/filter sizing against your pool volume for the recommended daily run time)
- Address any dead spots in circulation, sometimes requiring a return jet adjustment or, in some layouts, an added circulation aid
- Consider a phosphate remover if algae keeps recurring despite good chlorine practice; the working threshold most sources cite is treatment above roughly 1,000 ppb, with chlorine still the primary defense and phosphate control a secondary lever

The bottom line: Algae blooms overwhelmingly trace back to insufficient free chlorine relative to CYA sustained over time, not bad luck. Recovery requires aggressive shock dosing, continuous filtration, and daily brushing until FC holds steady rather than crashing, and preventing recurrence means maintaining the FC-to-CYA relationship consistently, not just reacting after the water already looks wrong.

Chapter 7: Troubleshooting Cloudy, Colored, and Problem Water

Problem: Cloudy (but not green) water

Step 1. Check FC, CC, pH, and TA; cloudiness commonly traces back to inadequate chlorine, poor filtration, or unbalanced water (particularly high pH or high TA causing minerals to fall out of solution).

Step 2. Check filter condition; a dirty, overdue-for-cleaning filter is one of the most common causes of persistent cloudiness even when chemistry looks reasonable.

Step 3. Shock the water if FC/CC readings suggest a chlorine deficit, and run the filter continuously until clarity improves.

Step 4. If cloudiness persists despite balanced chemistry and a clean filter, consider a clarifier or flocculant product (a typical clarifier dose is around 1 oz per 5,000 gallons, run the filter 8+ hours after dosing), or check for very high TDS as an underlying cause requiring partial drain-and-refill.

Problem: Green water (mild tint, not full algae swamp)

Treat as early-stage algae; see Chapter 6's recovery process, typically needing a less aggressive shock dose than a fully green swamp but the same fundamental sequence.

Problem: Cloudy white/gray water with a "milky" look

Step 1. Check calcium hardness and pH/TA together; this pattern often points to calcium or mineral scaling starting to precipitate out of solution, particularly if pH or TA are on the high end.

Step 2. Correct pH/TA toward the middle of their target ranges rather than the high end, and retest.

Problem: Brown/rust-colored staining or discoloration

Step 1. Check for metals in the water (iron, copper, manganese), sometimes from fill water source or from a metal-based algaecide used previously; a metal-specific test kit or a professional water analysis can confirm.

Step 2. If metals are confirmed, a metal sequestrant product, added per instructions, is the typical treatment; avoid shocking heavily before sequestering metals, since a strong oxidation event can actually precipitate dissolved metal and cause staining rather than prevent it.

Problem: Eye/skin irritation despite "chlorine present" on a test strip

Step 1. Check specifically for elevated combined chlorine (chloramines), not just total or free chlorine, since chloramines are the actual, well-documented irritant, not properly maintained free chlorine. Most "chlorine irritation" is really a chloramine problem — a chlorine-smelling pool is usually under-chlorinated, not over-chlorinated.

Step 2. Shock to clear combined chlorine, and check pH, since pH outside the comfortable 7.2–7.6 range compounds irritation.

Problem: Chlorine "disappears" within hours of adding it

- Step 1.** Check CYA; low CYA lets sunlight rapidly degrade chlorine, especially in an outdoor pool with strong sun exposure.
- Step 2.** If CYA is adequate but chlorine still vanishes fast, suspect high chlorine demand from an early algae bloom or heavy organic load, and shock accordingly rather than continuing routine small doses that get consumed as fast as they're added.

Problem: Foam on the water surface

- Step 1.** Usually linked to certain algaecides (particularly older or lower-quality quaternary ammonia-based products), lotions/sunscreen buildup, or in some cases low calcium hardness in soft-water areas.
- Step 2.** Check calcium hardness and consider an anti-foam product for a quick fix, alongside identifying and addressing the underlying source if it's a specific algaecide or product you've been using.

Problem: Hazy water during pollen season

Pollen grains are commonly too small to be reliably caught by a standard sand or cartridge filter on their own. Skim daily with a fine-mesh net (pollen collects overnight), run the filter longer or continuously during heavy pollen days, brush walls and waterline tile to knock loose settled pollen back into circulation, and dose a clarifier per label if the water stays hazy after skimming and filtering.

Quick reference table

Symptom	Likely cause	First step
Cloudy, not green	Low chlorine, dirty filter, or unbalanced pH/TA	Test and correct FC/pH/TA, clean filter
Mild green tint	Early algae	Shock per Chapter 6
Milky white/gray	Scaling from high pH/TA/calcium	Correct pH/TA toward mid-range
Rust/brown staining	Dissolved metals	Sequester before shocking heavily
Irritation despite "chlorine present"	High combined chlorine (chloramines)	Shock, recheck pH
Chlorine vanishing fast	Low CYA or high chlorine demand	Check CYA, consider shock if demand-related
Surface foam	Algaecide, lotions, or low calcium	Anti-foam product, check calcium

The bottom line: Most water clarity and comfort problems trace back to one of a small number of root causes: chlorine deficit relative to CYA, unbalanced pH/alkalinity, a dirty filter, or elevated

combined chlorine. Test before treating, since the visible symptom often points to more than one possible cause, and the wrong fix wastes both time and chemicals.

Chapter 8: Opening and Closing Your Pool

Opening for the season

1. **Remove and clean the winter cover**, letting debris fall away from the pool rather than into it.
2. **Reconnect and inspect equipment**: pump, filter, heater (if applicable), and check for any winter damage to pipes, fittings, or the liner/surface.
3. **Bring water level up** to the normal operating level if it dropped over winter.
4. **Prime and start the pump**, checking for proper flow and no unusual noises.
5. **Test water thoroughly**: FC, CC, pH, TA, CYA, calcium hardness, since water sitting all winter, even with a cover, typically needs a full rebalance.
6. **Balance in the standard order** (Chapter 5): alkalinity, pH, calcium, CYA if needed.
7. **Shock the pool**, typically at a higher-than-routine dose for opening, since winter water usually carries some organic buildup even under a well-fitted cover.
8. **Run the filter continuously for the first day or two**, brushing surfaces to help clear any early growth that started during closure.
9. **Retest after 24–48 hours** and resume normal maintenance dosing once FC holds steady and water is clear.

Closing for the season

1. **Balance the water fully** (Chapter 5) before closing; closing with unbalanced water invites scaling, staining, or corrosion issues that show up as an unpleasant surprise at next season's opening.
2. **Shock the pool** a few days before closing, giving combined chlorine time to clear and residual chlorine time to settle to a stable, appropriate level for the closed period.
3. **Add a winterizing algaecide** if closing in a climate where the pool will sit dormant for an extended period, following product-specific guidance for your pool type and climate.
4. **Lower water level appropriately** for your specific pool type and climate (this varies significantly: some pools are lowered below skimmer level for freeze protection, others are not; follow guidance specific to your pool type, plumbing configuration, and climate).
5. **Blow out or otherwise winterize plumbing lines** in freeze-risk climates, adding appropriate antifreeze to lines per your pool's specific winterizing procedure, to prevent freeze damage to pipes and equipment.
6. **Clean and store removable equipment** (ladders, skimmer baskets, etc.) appropriately, and service or store the filter per manufacturer guidance for the off-season.
7. **Install a properly fitted winter cover**, checking it's secure and won't collect standing water or excessive debris through the off-season, both of which strain the cover and can affect spring water quality.

Regional variation

Closing procedures vary significantly between climates that see hard freezes (requiring full winterization of plumbing) and mild climates where pools may run year-round or close only briefly (requiring less aggressive winterization but still benefiting from a proper balance-and-shock before an extended low-use period). If you're unsure what your specific climate and pool type require, a local pool professional familiar with your region's typical opening/closing procedures is worth a consultation, particularly for your first season with a new pool.

The bottom line: Both opening and closing revolve around the same core principle: balance the water fully and shock it before a period of reduced attention (winter dormancy, or the gap between closing and the following season's opening). Skipping a proper balance-and-shock before closing is one of the most common causes of a difficult, algae-plagued opening the following spring.

Chapter 9: Salt Systems and Alternative Sanitizers

How salt chlorine generators work

A salt chlorine generator (salt water pool) doesn't eliminate chlorine — it generates it on-site. Dissolved salt in the water passes through a generator cell that uses electrolysis to convert it into chlorine, which then sanitizes the water exactly as chlorine from any other source would. A salt pool is still a chlorine pool; it's simply generating its own chlorine continuously instead of you adding it manually each time.

What's different about managing a salt pool

- **Salt level itself needs periodic testing** (typically a few times per season), since the generator needs an adequate salt concentration to function properly — commonly 2,700–3,400 ppm, with about 3,200 ppm the most widely cited universal target. Below 2,700 ppm the cell can't generate enough chlorine; above 3,400 ppm many generators self-protect and shut down.
- **CYA still matters just as much**, since salt-generated chlorine is still degraded by sunlight the same way any free chlorine is; salt pools often run CYA on the higher end of the acceptable range (60–80 ppm) specifically to protect the continuously generated chlorine.
- **The generator cell needs periodic cleaning** to remove calcium scale buildup that naturally forms on the cell's plates over time, following manufacturer-specific cleaning guidance and schedule.
- **You'll still need to shock periodically**, since the generator's continuous output usually isn't enough to catch up quickly after heavy bather load or an algae scare; most systems include a "boost" or "super chlorinate" mode for this, or you can add chlorine manually.
- **Salt cells have a finite lifespan** — commonly 3–7 years depending on usage and maintenance, with typical replacement cost roughly \$700–\$1,100 (some models as low as \$300–\$500) — a real eventual replacement cost worth budgeting for. Combined annual running cost (salt top-off plus added electricity, excluding amortized cell replacement) runs roughly \$80–\$250/year.

Other alternative sanitizing systems

- **Ozone and UV-C systems:** supplemental secondary sanitizers/oxidizers, not chlorine replacements. A properly sized system can reduce required chlorine use by roughly 50–90% versus a chlorine-only pool, but neither leaves a residual sanitizer in the water, so a base level of chlorine or bromine is still required between passes through the unit.
- **Bromine:** more common in spas than pools due to cost, but a viable alternative sanitizer. Bromine produces bromamines instead of chloramines when it reacts with contaminants, which are generally less irritating, though bromine itself can still cause a rash or itch at high levels, and true bromine allergy exists. Requires its own specific balancing targets, somewhat different from chlorine's.
- **PHMB (polyhexamethylene biguanide):** a non-chlorine, non-bromine sanitizer marketed as hypoallergenic; it requires its own dedicated hydrogen-peroxide-based shock and is not compatible with chlorine, meaning a pool can't casually switch back and forth between the two systems.

Choosing a system

Salt chlorine generation is the most common alternative for residential pools, largely because it reduces the manual labor of daily/weekly chlorine dosing while still being, fundamentally, chlorine sanitization with familiar chemistry. Ozone and UV systems appeal to owners wanting to minimize total chlorine use and are often chosen alongside a base chlorine or salt system rather than in place of one. Bromine suits pools (more commonly spas) where the specific properties of chlorine odor are a bigger concern than bromine's higher typical cost.

The bottom line: A salt pool is still a chlorine pool, generated on-site rather than added manually, and it still requires the same fundamental chemistry management (FC, pH, alkalinity, CYA) plus salt-specific testing and generator cell maintenance. Ozone and UV systems typically supplement rather than replace a base chlorine or salt system in most residential setups.

Chapter 10: Equipment Care and Common Repairs

The pump

Your pump circulates water through the filter and back to the pool; without adequate daily run time, even perfectly dosed chemicals won't distribute evenly or filter effectively. Run time needed depends on your pump's flow rate relative to your pool's volume; a common rule of thumb targets turning over the pool's full volume at least once per day, though bigger pools, heavy bather load, or algae recovery (Chapter 6) may call for longer run times. Listen for unusual noise (a failing bearing often sounds like a grinding or whining change from normal) and check for proper prime (air in the system, low flow, or an air leak on the suction side are common causes of a pump losing prime).

The filter

Three common filter types, each with its own maintenance pattern:

- **Sand filters:** backwashed periodically (reversing water flow to flush trapped debris out to waste) based on a pressure gauge reading rising a set amount above the clean, freshly backwashed baseline; sand media typically needs full replacement every several years.
- **Cartridge filters:** removed and hosed off periodically, with a deeper chemical soak/clean a few times per season, and full cartridge replacement every few seasons depending on use and condition.
- **DE (diatomaceous earth) filters:** backwashed similarly to sand filters but require adding fresh DE powder after each backwash; offer the finest filtration of the three common types but require the most hands-on maintenance.

Common equipment issues

- **Low water flow/pressure:** check for a clogged skimmer or pump basket first (the most common and easiest fix), then a dirty filter needing cleaning/backwashing, then check for a suction-side air leak if priming issues persist.
- **Pump losing prime repeatedly:** usually an air leak on the suction side (a loose fitting, a cracked lid gasket, or low water level exposing the skimmer to air); check water level and visible fittings before assuming a more serious pump issue.
- **Heater not heating:** check the gas supply/pilot (for gas heaters) or breaker (for electric/heat pump units) first, then check for adequate water flow, since most heaters have a safety flow switch that prevents operation without sufficient circulation.
- **Automatic pool cleaner not moving properly or losing suction:** check the cleaner's own filter bag/basket for a clog, and check hose connections for leaks reducing suction, before assuming the cleaner itself has failed.

When to call a professional

- Any electrical work on pumps, heaters, or lighting, unless you're specifically qualified
- Suspected leaks in underground plumbing (a consistently, unexplainably dropping water level beyond normal evaporation is the classic sign, and pinpointing an underground leak generally requires specialized equipment)
- Structural issues with the pool surface (cracking, significant staining that won't respond to sequestering, or surface delamination)
- Heater repairs involving gas lines or refrigerant (for heat pump units)
- Salt cell replacement or generator troubleshooting beyond basic cleaning, if you're not comfortable with the specific unit's procedures

A simple evaporation check for suspected leaks

Fill a bucket with pool water, set it on a pool step so the water level inside and outside the bucket match, mark both levels, and check after 24 hours. If the pool level has dropped noticeably more than the bucket's, that's a reasonable (though not definitive) indicator of a leak beyond normal evaporation, worth a professional leak detection if it persists.

The bottom line: Most common equipment problems (low flow, priming issues, a struggling cleaner) trace back to a clogged basket, dirty filter, or air leak, and are reasonable for an owner to check before assuming a major failure. Electrical work, suspected underground leaks, and major structural or heater issues are professional territory.

Chapter 11: Costs, Budgeting, and Building Good Habits

Typical seasonal cost ranges (2026 estimates; residential in-ground pool, varies by region and pool size)

Item	Typical range
Routine chemicals (full season)	\$300–\$800
Test kit (liquid reagent, replaced periodically)	\$30–\$80
Test strips (ongoing supply)	\$15–\$40 per season
Professional opening service	\$150–\$400
Professional closing service	\$200–\$450
Weekly professional service (full season, if outsourced)	\$1,200–\$3,500+
Filter cartridge replacement	\$50–\$200 per cartridge, varies by system
Sand filter media replacement	\$100–\$300, every several years
Pump replacement	\$500–\$1,500 installed
Salt cell replacement	\$400–\$1,000
Heater repair	\$200–\$800 for common repairs; more for major component failure
Algae recovery (chemicals only, DIY)	\$50–\$200, depending on severity
Professional algae recovery service	\$200–\$600+
Resurfacing (plaster pools, periodic)	\$4,000–\$15,000+, every 10-20 years depending on surface type

DIY vs. professional service

Weekly maintenance, basic testing, and routine dosing are well within reach for most owners willing to spend 30–60 minutes a week on it, and this is where the real cost savings live compared to a full weekly service contract. Opening, closing, and more complex repairs are where many owners choose to bring in a professional, particularly in climates with real winterization requirements, or when equipment troubleshooting goes beyond the basics in Chapter 10.

Preventing the expensive mistakes

The costliest pool-ownership mistakes are almost always preventable with consistent basic habits: letting FC run chronically low relative to CYA (leading to algae recovery costs), closing with unbalanced water

(leading to scaling, staining, or a rough spring opening), ignoring a slowly dropping water level (a small leak becomes a bigger, more expensive one the longer it runs undetected), and skipping filter maintenance (reducing equipment lifespan and water clarity).

Building a seasonal budget

A practical approach: budget your routine chemical and testing costs as a predictable seasonal line item, and set aside a modest reserve for an eventual equipment repair or replacement, since pumps, filters, heaters, and salt cells all have finite lifespans and it's a matter of when, not if, one needs attention. Pair that with the habits in this book, and most seasons will fall into the cheap, predictable maintenance tier rather than the expensive, reactive repair tier.

The bottom line: Routine chemicals and testing cost a few hundred dollars a season; algae recovery and equipment repairs cost hundreds to a few thousand; major resurfacing or leak repair can run into five figures. Consistent basic habits — testing regularly, balancing before chlorine, brushing weekly, and closing properly — are what keep most pool owners in the cheap tier for the vast majority of their ownership.

Appendix A: Printable Water Test Log

Date	FC	CC	pH	TA	CYA	Calcium	Chemicals added	Notes

Pool reference information (fill in once):

Field	Your pool's info
Pool volume (gallons)	
Surface type (plaster/vinyl/fiberglass)	
Sanitizer type (chlorine/salt/bromine/other)	
Filter type (sand/cartridge/DE)	
Pump model	
Heater (Y/N, type)	

Appendix B: Glossary of Terms

Alkalinity (TA): A measure of water's buffering capacity against pH change; low alkalinity causes unstable, swinging pH.

Chloramine: A compound formed when chlorine reacts with contaminants (sweat, urine, organics); the primary source of "chlorine smell" and eye irritation, measured as combined chlorine.

Combined chlorine (CC): Chlorine already reacted with contaminants; largely ineffective as a sanitizer and irritating to swimmers at elevated levels.

Cyanuric acid (CYA): Also called stabilizer or conditioner; protects free chlorine from rapid UV degradation but reduces chlorine's effective killing power at higher concentrations.

Free chlorine (FC): The active, available chlorine capable of sanitizing the water; the most important number to track.

Salt chlorine generator: Equipment that converts dissolved salt into chlorine via electrolysis, generating sanitizer on-site rather than requiring manual addition.

Sequestrant: A chemical that binds dissolved metals in water to prevent staining, used before heavy shock treatments when metals are present.

Shock: A large dose of chlorine or oxidizer used to rapidly clear combined chlorine, organic material, or algae; a periodic maintenance action, not routine daily dosing.

Total dissolved solids (TDS): A measure of all dissolved substances in pool water; rises naturally over a season and can make water harder to balance at very high levels.

Turnover rate: The time it takes a pool's filtration system to circulate a volume of water equal to the entire pool once.

Appendix C: 20 Questions to Ask Before You Hire a Pool Service Company

1. Are you licensed and/or insured to perform pool service in this state?
2. What's included in your standard weekly visit (testing, brushing, skimming, vacuuming, chemical balancing)?
3. Will you provide a written record of test results and chemicals added at each visit?
4. How do you handle an emerging algae problem — is it included in standard service or billed separately?
5. What's your typical response time for an urgent issue (equipment failure, sudden cloudy/green water)?
6. Do you offer opening and closing services, and what's included in each?
7. What brands/types of chemicals do you use, and can I ask you to avoid or use specific products?
8. How do you handle equipment repairs — in-house or subcontracted?
9. Will you provide a written estimate before any repair beyond routine chemical service?
10. What's your policy if I'm unhappy with water clarity or balance between scheduled visits?
11. Do you carry liability insurance, and can you provide proof?
12. How long have you served this specific area, and are you familiar with our local water source/fill-water characteristics?
13. Can you provide references from current customers with a similar pool type and size?
14. What's your cancellation or pause policy for off-season months?
15. Do you offer a service guarantee or satisfaction policy?
16. How do you handle salt cell or specialty equipment maintenance if I have a salt system?
17. What would trigger an additional charge beyond the standard weekly rate?
18. Can I get a same-season inspection before committing to a full-season contract?
19. How do you communicate results and recommendations — an app, a written note, a phone call?
20. If you find equipment needing repair beyond your scope, do you handle it or refer me elsewhere?

Closing Note

Pool ownership isn't difficult once you know the handful of things that matter: understand free chlorine versus combined chlorine, balance in the right order, maintain the FC-to-CYA relationship consistently, and brush every week whether the water looks like it needs it or not. Keep this book somewhere you'll actually find it again, whether that's a shelf, a saved file, or a printed copy in the pool shed.

For the free pool chemical calculator, seasonal reminders, and further guides, visit **clearcovepools.com/guides**.

This book provides general educational guidance based on standard industry dosing practices and CDC/MAHC health-authority ranges. It is not a substitute for your own test kit results, product-specific label instructions, a product's Safety Data Sheet, or advice from a licensed pool professional familiar with your specific pool, surface, and local water conditions.

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